MSG No. 120.324 $I\bar{4}'c2'$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[3,4,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[11,12,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,7,8]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[11,12,13,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,15,16]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[3,4,13,14]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[7,8,9,10]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,13,14]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4,15,16]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[7,8,11,12]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,5]`
2	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[2,6]`
3	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[3,8]`
4	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[4,7]`
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[9,13]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[10,14]`
7	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[11,16]`

8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[12,15]

Table 6: Wyckoff site: 8f, site symmetry: 2..

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,2]
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[3,4]
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[5,6]
4	$[0,\ 0,\ -z]$	[7,8]
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[9,10]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[11,12]
7	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[13,14]
8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[15,16]

Table 7: Wyckoff site: 8g, site symmetry: 2..

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,2]
2	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[3,4]
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[5,6]
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	[7,8]
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[9,10]
6	$\left[\frac{1}{2},\ 0,\ z\right]$	[11,12]
7	$\left[0,\ \frac{1}{2},\ -z\right]$	[13,14]
8	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[15,16]

Table 8: Wyckoff site: 8h, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	[1,13]
2	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	[2,14]
3	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[3,16]
4	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[4,15]
5	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[5,9]
6	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	[6,10]
7	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	[7,12]
8	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	[8,11]

Table 9: Wyckoff site: 16i, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[3]

continued ...

Table 9

No.	position	mapping
4	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[4]
5	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[5]
6	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[6]
7	$[y,\, -x,\, -z]$	[7]
8	$[-y,\, x,\, -z]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[10]
11	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[11]
12	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[12]
13	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[13]
14	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[14]
15	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[15]
16	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[16]